

1. I originally tried to make the diffusion directional but I couldn't figure it out. Instead I decided to go for a style map approach. I thought about how to nicely add an interface for modifying the parameters of the simulation based on a few greyscale maps. I then thought back to a slime mold simulation I did at one point and remembered how I had each parameter have a "base" and then a "modifier". I could then do a classic $val = modifier * style_val + base$ to map the style value to a modified simulation input parameter. I then decided to use the gradient noise implementation from the Book of Shaders as the style and layered them similar to fractal noise (each having a different frequency). I also realized that I could make the modifier values negative too. I didn't realize this initially. My slime mold simulation didn't work with sliders. They were unbounded. I then made the minimum values negative. Then I just had a lot of fun messing around with it. WebGPU is so much faster on Firefox its so funny. It also breaks everything so much more because of incredibly weird and unexpected things. I originally wanted to make the simulation be black patterns on white and luckily I didn't need to make any changes because its like that by default.